A BRIEF REPORT ON

SATELLITE AND COMMUNICATION GROUND STATION

AND

RF EXPERIMENTATION PLATFORM

Submitted by

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1. Abstract

This report covers our work on estabilishing a student run ground station to receive and process satellite signals using VHF antennas, Software Defined Radios, tracking software, signal-processing tools and decoding tools.The ground station is currently under development, and the experiments reported here represent the initial stages of establishing a functional reception and processing chain. Till date We built and tested both V-dipole and Quadrifilar Helix antennas around 137 MHz, achieving a clear 71 dB SNR (maximum) signal in close-range tests, though we noticed some noise and latency during two-way testing. We set up our software pipeline using GNU Radio - for transmission of Live Audio, GPredict, SDR Console and SatDump alongside SDR hardware like the HackRF, Pluto SDR, and USRP B200 Mini, and successfully verified our decoding workflow offline using a pre-recorded NOAA satellite file available on SatNogs Observation with the help of WXtoImage. Overall, while the initial hardware and software setup works in concept, we are currently working on fixing noise issues, tuning the antennas, and achieving our first reliable live satellite reception.

2. Introduction

A satellite ground station receives and processes data signals from space, handling tasks like frequency tuning, satellite tracking, demodulation, and data visualization. Building a small-scale ground station at the university level provides us with a practical, hands-on platform to understand radio-frequency (RF) engineering, software-defined radios (SDRs), antenna design, and digital signal processing. To keep the development manageable, we chose a modular architecture instead of a single fixed system. This allows us to test, troubleshoot, and characterize individual antennas, SDR hardware platforms, and software tools independently before integrating them into an end-to-end processing pipeline.

Our current focus is on the 137 MHz VHF band, which is standard for weather satellite transmissions. So far, we have designed and tested a V-dipole antenna, built a Quadrifilar Helix (QFH) antenna as an alternative setup, and performed controlled local transmit-receive tests between them to evaluate their practical RF performance. Alongside hardware testing, we established our software and tracking workflow using tools like GNU Radio, GPredict, and WXtoImg. To verify that our decoding chain works, we successfully processed a pre-recorded NOAA satellite audio file offline using WXtoImg. While this lab test confirmed our software workflow, capturing and decoding a reliable live satellite pass remains our main objective as development continues.

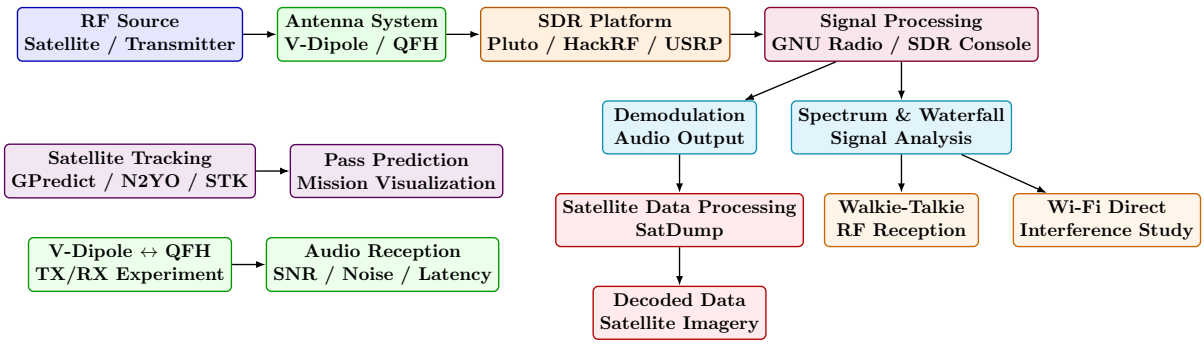
3. Objectives

The major objectives of the project are:

1. To develop a student-scale ground station.
2. To design, construct, and test suitable antennas, currently,done a V-dipole and QFH antenna.
3. To interface the antennas with software-defined radio hardware for RF signal acquisition.
4. To investigate signal reception and audio recovery under controlled experimental conditions.
5. To evaluate the V-dipole and QFH as a transmit–receive pair in both directions.
6. To use satellite tracking software for identifying satellite passes and corresponding operating frequencies.
7. To establish a signal-processing and decoding workflow using tools such as GNU Radio and SatDump.

4. Ground Station Architecture

The developed ground-station platform is designed as a modular RF experimentation system rather than being restricted to a single satellite communication chain. The platform combines antenna systems, software-defined radios, signal-processing software, tracking tools, and controlled RF communication experiments. The overall experimental framework can be represented as follows:



Our setup is flexible enough to test a few different RF setups besides just satellites. For satellite work, we targeted the 137 MHz VHF band using V-dipole and QFH antennas, using SDRs like the Pluto SDR, HackRF, and USRP B200 Mini to bridge the hardware to our digital tools. We also used the platform for ground-level tests, like receiving walkie-talkie signals to look at real-world spectrum performance, and running a controlled test to see how Wi-Fi Direct interferes with a signal link. Driven by Open Source software like GNU Radio, SDR Console, SatDump, WXtoImg, GPredict, N2YO, and STK, the modular setup let us easily swap antennas and SDR platforms depending on what we were testing.

5. Hardware Components

5.1. V-Dipole Antenna

The V-dipole was our main starting antenna because its simple design makes it easy to build, adjust, and test in the lab. In a close-range test, it successfully picked up audio with an observed SNR of around 71 dB. Since SNR varies with distance, power, gain, and environment, this figure is just a test measurement rather than a fixed spec. Overall, it proved to be a reliable, easy-to-use option for our initial VHF reception experiments.



Figure 1: V-dipole Antenna

5.2. QFH Antenna

We built a Quadrifilar Helix (QFH) antenna because its circular polarization makes it better suited for satellite passes than a basic dipole. Beyond passive testing, we ran two-way transmit-receive tests between the QFH and V-dipole, swapping which one transmitted and which received. Audio was successfully passed in both directions, proving they can form a functional RF link. However, we ran into noise and latency issues that we still need to fix.

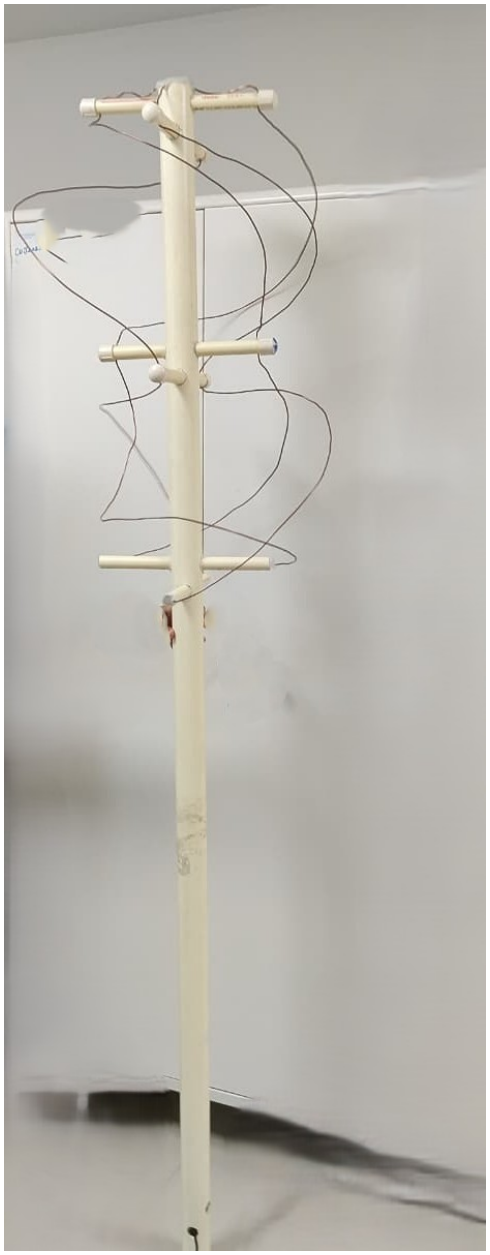


Figure 2: QFH Antenna

5.3. Software Defined Radios

Software Defined Radios (SDRs) serve as the bridge between our antennas and software by converting incoming RF signals into digital data. We tested three different SDR platforms: the Pluto SDR, HackRF, and USRP B200 Mini. Their tunable frequency

range allowed us to test and compare how our antenna setup performs across various software environments.

6. Software Tools

Table 1: Software tools used in the ground station development

Software	Purpose in the project
GNU Radio	Block-based RF signal processing, filtering, demodulation, and signal-flow experimentation.
SDR Console	SDR control, frequency tuning, spectrum/waterfall observation, and audio monitoring.
GPredict	Satellite tracking and pass prediction.
N2YO	Online satellite tracking and pass information.
STK	Satellite orbit/mission visualization and analysis.
SatDump	Satellite signal processing, decoding, and data/image generation where applicable.

6.1. GNU Radio

GNU Radio is our main tool for signal processing and generation. It lets us build custom pipelines for filtering, demodulating, and handling audio, while also letting us create test tones to check our setup. Because it hooks straight into the SDR, we can adjust our software pipeline directly without changing the physical hardware. This makes it much easier to test different processing setups in software before dealing with real-world signals.

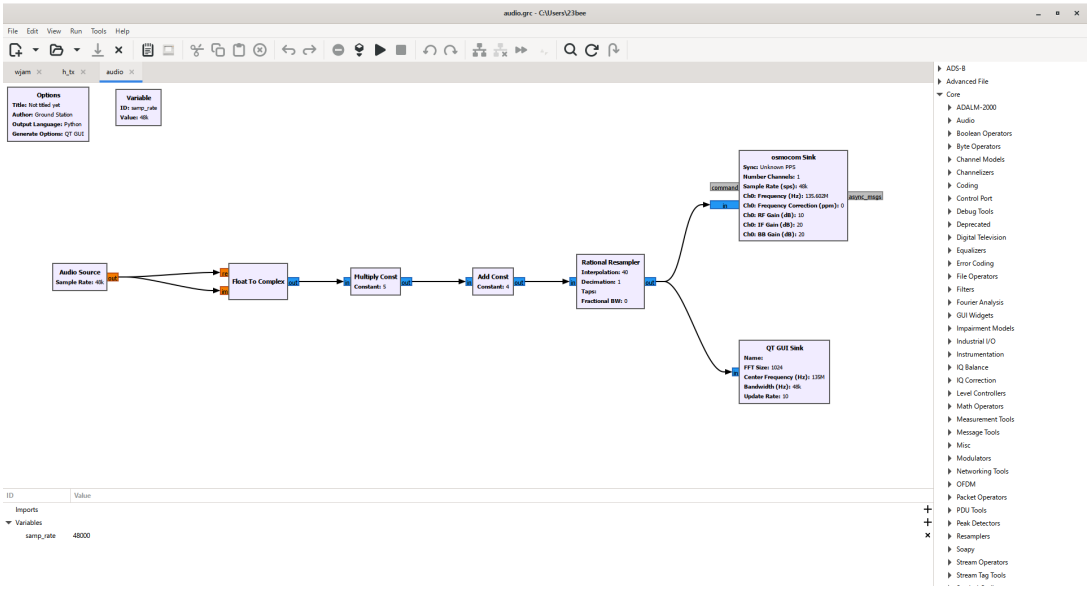


Figure 3: GNU Radio flowgraph used during signal-processing experiments

6.2. GPredict

GPredict handles our satellite tracking and pass predictions. It tells us when a satellite will be overhead and how long our reception window will last. Having this tracking data is essential because we only get a few minutes per pass, and the relative motion causes a Doppler shift that shifts our target frequency while the satellite moves.

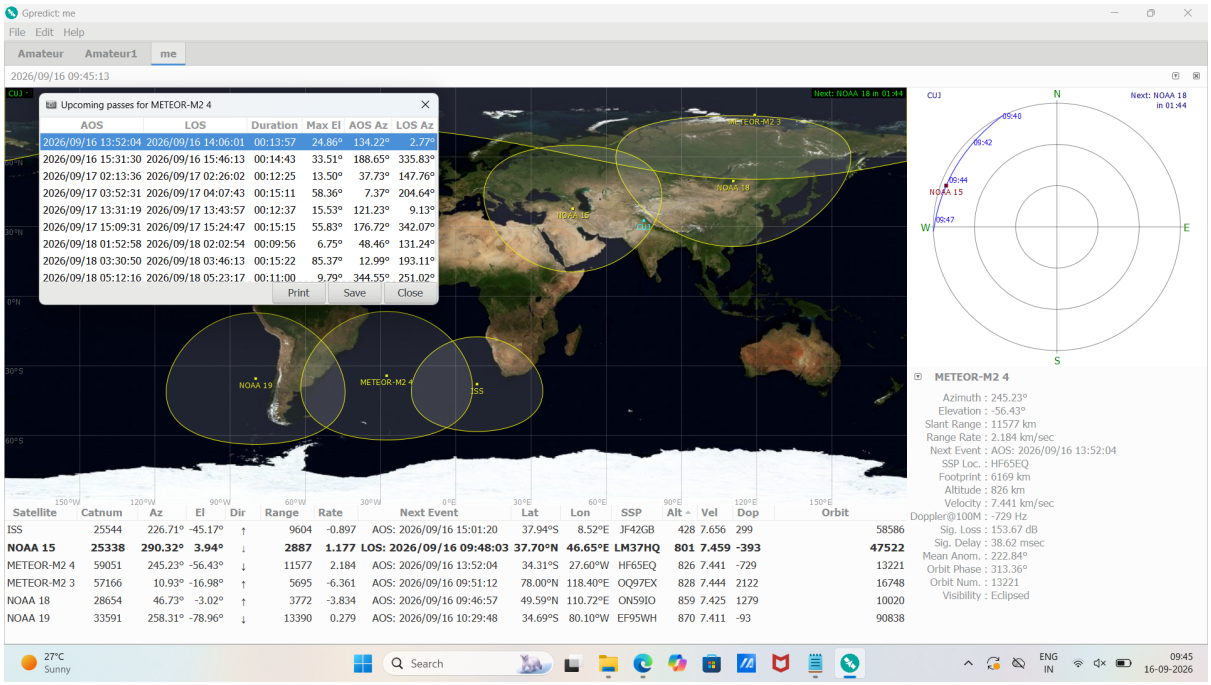


Figure 4: GPredict satellite tracking/pass-prediction screen used for planning observations.

6.3. SDR Console

SDR Console serves as our main software interface for controlling and monitoring the SDR. It gives us a real-time spectrum and waterfall display, making it easy to spot signal spikes around our tuned frequency visually. We also use it to monitor and listen to audio live during our reception tests.

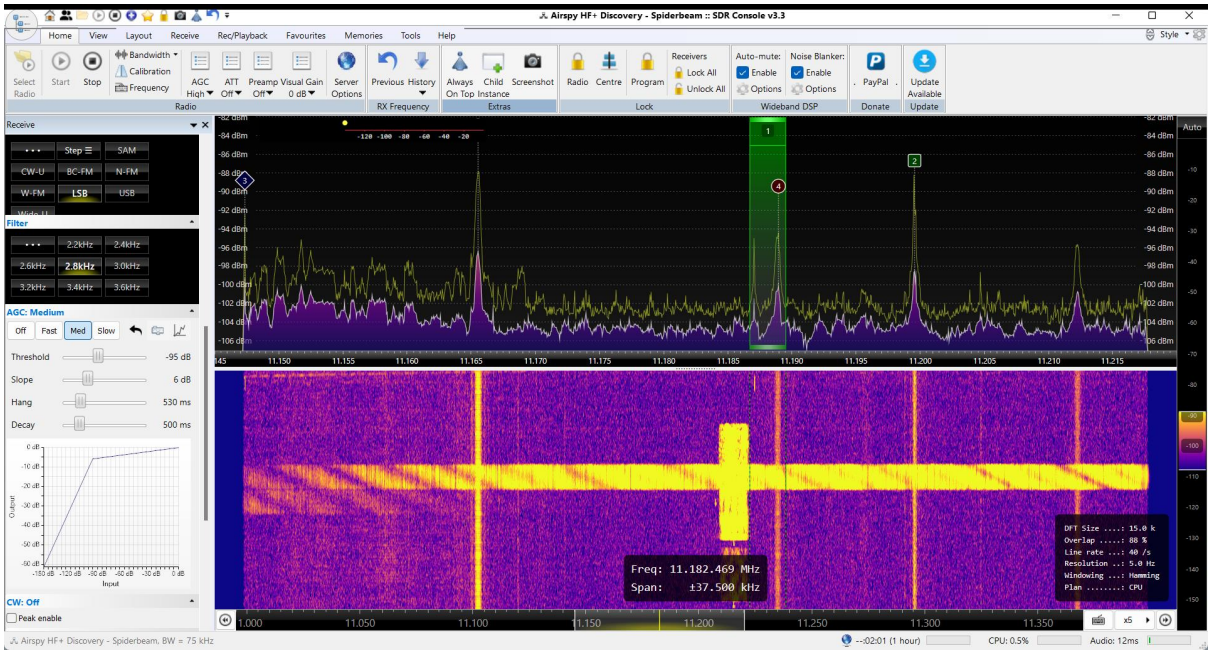


Figure 5: SDR Console spectrum/waterfall or audio-reception screen.

6.4. SatDump

SatDump is our main tool for processing and decoding satellite signals, handling everything from demodulation to extracting the final data based on the satellite's protocol. In our project, we used it to run an offline test on a pre-recorded audio file to verify that our processing workflow actually works. This was strictly a dry run to test the software setup, not a live satellite capture.



Figure 6: SatDump processing/decoding interface or output obtained from the offline test

7. Experimental Methodology

The experimental work was divided into several stages so that antenna performance, RF transmission/reception, satellite tracking, and decoding could be evaluated separately.

7.1. V-Dipole Audio Reception Test

We started by connecting the V-dipole to our receiver setup and testing audio reception over a short distance. The goal here was simple: make sure the whole pipeline works, from the antenna through the SDR down to the final audio output.

In the close-range lab setup, the antenna successfully picked up the signal, giving us a clean audio output with an observed SNR of around 71 dB. To see how it held up outside a controlled environment, we expanded our testing by moving to a much longer distance and rotating the antennas through different angles to check off-axis performance. As expected, the signal-to-noise ratio dropped compared to our initial 71 dB test, but the audio remained clear and intelligible. These tests proved that our basic receiving chain works and gave us a clearer picture of how signal strength degrades with distance and alignment shifts before we move on to live satellite passes.

7.2. QFH and V-Dipole Bidirectional Experiment

A second experiment investigated the two antennas as a controlled transmit–receive pair. Two configurations were tested:

1. **Configuration A:** QFH used as the transmitter and V-dipole used as the receiver.
2. **Configuration B:** V-dipole used as the transmitter and QFH used as the receiver.

Audio was received both ways, confirming a functional RF link in either configuration. However, the signal had noticeable noise and latency, likely caused by SDR buffering, software processing delays, improper gain settings, or environmental RF noise. We need further testing to isolate and fix the exact sources of these delays and noise.

Table 2: Summary of antenna-pair experiments

Transmitter	Receiver	Audio observed	Observation
QFH	V-dipole	Yes	Noise and latency observed
V-dipole	QFH	Yes	Noise and latency observed

7.3. VHF Frequency Observation

Our ground station development focuses on the VHF band around 135-138 MHz. During testing, we mostly operated around 135.6 MHz, along with some observations around 137 MHz.

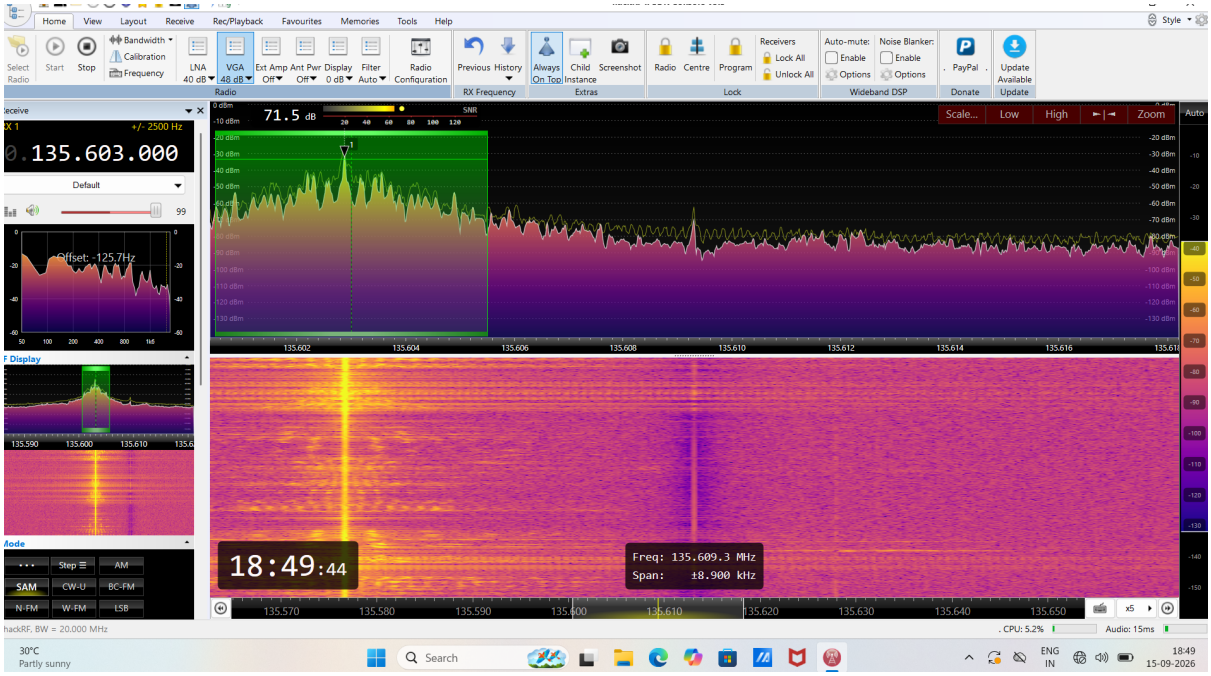


Figure 7: spectrum/waterfall observation around the 135.6 MHz region.

7.4. Satellite Tracking

Satellite tracking bridges our orbit predictions with the actual RF setup. We use GPredict and N2YO to see when a satellite is overhead and plan our reception timing, while STK helps us picture its path across the sky. For future live tests, this info lets us set our SDR's frequency, sample rate, gain, and recording schedule ahead of time, right before the satellite shows up over the horizon.

8. NOAA Offline Decoding Trial

We ran an offline decoding test using a pre-recorded NOAA audio file to check our software pipeline without needing a live satellite pass. This proved our decoding setup actually works, though it's important to clarify that this was a dry run with existing data, not a live satellite capture. The next step is to run this exact same workflow on a live signal captured directly through our own antenna and SDR setup. Doing that will give us complete end-to-end validation, from the physical hardware all the way to the decoded imagery.

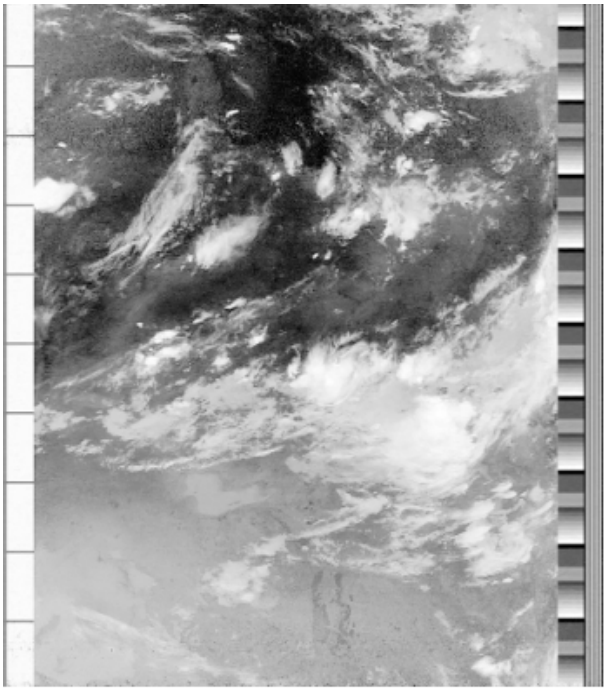


Figure 8: Output obtained during the offline NOAA decoding trial using a pre-recorded audio file.

9. Results and Observations

The initial experiments produced the following observations:

1. The V-dipole was able to receive an audio signal during a close-range controlled experiment.
2. An SNR of approximately 71 dB was observed during the particular V-dipole test.
3. The QFH antenna was successfully included in the experimental setup and tested with the V-dipole.
4. Audio was received when the QFH was used as transmitter and the V-dipole as receiver and vice-versa.
5. Noise and latency were present in the bidirectional audio experiments.
6. The software chain for satellite tracking and signal processing has been established using tools including GPredict, GNU Radio, SDR Console, and SatDump.
7. The project has investigated the VHF region around 135-138 MHz, including observations around 137.9 MHz and 137.1 MHz.
8. The offline NOAA decoding experiment successfully provided a method for testing the decoding workflow using a pre-recorded audio file.

Table 3: Current experimental status

Activity	**Current status**
V-dipole audio reception	Successfully demonstrated in close-range test
Observed SNR	Approximately 71 dB in the reported test
QFH testing	Conducted
QFH TX → V-dipole RX	Audio received; noise/latency observed
V-dipole TX → QFH RX	Audio received; noise/latency observed
Satellite tracking workflow	Established for further experiments
Offline NOAA decoding	Tested using pre-recorded audio
Live satellite reception	Ongoing development

10. Challenges and Limitations

The ground station is currently under development, and several technical challenges remain before it can be considered a fully operational satellite receiving station.

10.1. Noise

Noise was present during our antenna-pair tests, which can come from several sources—including local RF interference, SDR gain settings, receiver bandwidth, cable loss, impedance mismatch, and antenna alignment. To keep our data consistent and repeatable, future tests need to log key parameters like receiver gain, overall bandwidth, exact distance, orientation, and measurement bandwidth.

10.2. Latency

Latency was present in the received audio, likely caused by software buffering and audio-processing delays rather than the RF link itself. To fix this, future tests will systematically measure latency across different sample rates, buffer sizes, and software configurations to pinpoint the bottleneck.

10.3. Live Satellite Reception

Our local tests prove our basic antenna and receiver pipeline works, but a short-range audio link is very different from capturing a satellite signal. To achieve a complete satellite demonstration, we still need to handle a live pass, compensate for Doppler shift, align for proper polarization, maintain a sufficient link margin, and decode a signal recorded directly by our own ground station.

10.4. Antenna Optimization

Both the V-dipole and QFH need more testing to fully understand their real-world performance. Key factors like element dimensions, impedance matching, orientation, polarization, and local surroundings all impact how well they pick up signals. Further testing at our target 137 MHz band will help us decide which antenna configuration works best for the ground station.

11. Future Scope

The following activities are planned as the ground station develops further:

1. Complete reliable live satellite signal reception using the ground-station antenna system.
2. Improve the V-dipole and QFH designs through systematic measurements and tuning.
3. Characterize antenna performance at the intended VHF operating frequencies.
4. Investigate and reduce the observed audio latency through optimization of SDR and software buffering.
5. Improve noise performance through appropriate filtering, gain control, grounding, shielding, and RF setup optimization.
6. Integrate satellite pass prediction with the SDR reception workflow.
7. Investigate Doppler compensation during satellite passes.
8. Capture live satellite recordings and process them through SatDump or other appropriate decoders.
9. Generate and archive decoded satellite images/data for future analysis.
10. Develop a more automated reception workflow for scheduled satellite passes.

12. Conclusion

This project focuses on building a modular, student-led satellite ground station by integrating custom antennas, SDR hardware, and tracking and processing tools. Local tests confirmed our basic RF pipeline—both the V-dipole and QFH successfully established functional audio links, achieving up to 71 dB SNR in close-range tests, while highlighting noise and latency as key areas for optimization.

Our software chain (GNU Radio, SDR Console, GPredict, N2YO, STK, and SatDump) is fully integrated, and offline tests with pre-recorded NOAA files have successfully validated our signal decoding workflow. The next major milestone is capturing, demodulating, and decoding a live satellite signal end-to-end using our own hardware setup..

Acknowledgement

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